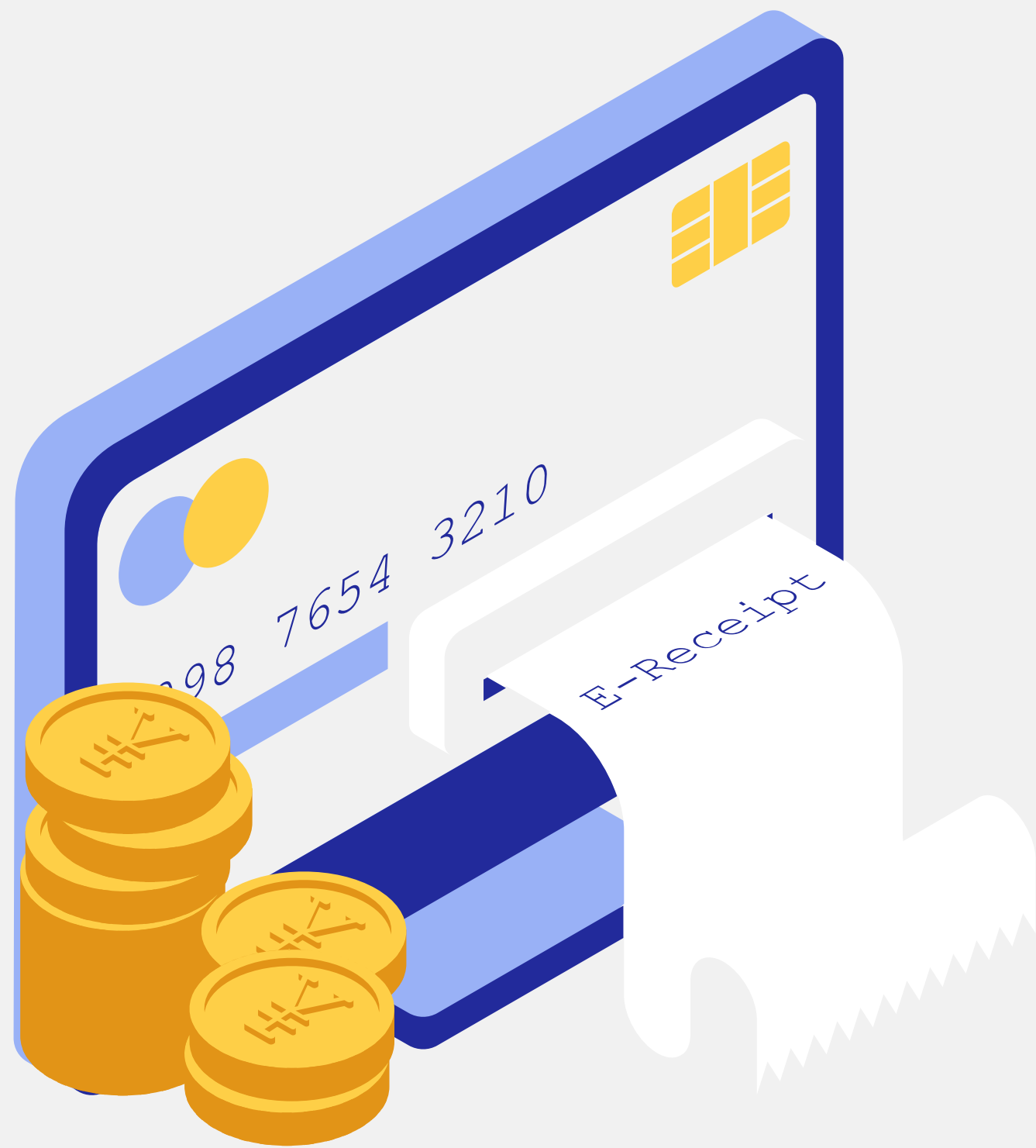




# Bank Customer Churn Analysis

**Bank Customer Retention Analysis using Microsoft Excel**

**Prepared by :  
Saranya Koduru**





# Business Background

ABC Bank has experienced a rise in customer churn, resulting in reduced revenue, increased customer acquisition costs, and lower long-term profitability. Management requires a data-driven understanding of:

- Which customers are leaving
- Why they are leaving
- How churn can be reduced through targeted retention strategies

## Objective

Analyze customer demographic and account data to identify the key drivers of customer churn and provide actionable business recommendations that improve customer retention.



# Business Questions

1. Which customers are most likely to churn?
2. Which customer characteristics influence churn?
3. Does geography affect churn?
4. Does age influence churn?
5. Does account balance affect churn?
6. Does the number of products impact churn?
7. Are inactive customers more likely to leave?
8. Does owning a credit card reduce churn?
9. Does customer tenure influence churn?
10. What actions can reduce customer churn?



# Dataset Overview

| Attribute          | Details                                       |
|--------------------|-----------------------------------------------|
| Domain             | Banking                                       |
| Dataset Source     | Maven Analytics – Bank Customer Churn Dataset |
| Dataset Size       | 10,000 Customer Records                       |
| Target Variable    | Exited (Customer Churn)                       |
| Integrated Dataset | Customer_Churn_Master                         |
| Analysis Tool      | Microsoft Excel                               |

# Methodology

## Tools Used

- Microsoft Excel
- Power Query
- Pivot Tables
- Pivot Charts
- XLOOKUP
- Slicers

## Analysis Techniques

- Data Cleaning & Validation
- Data Transformation
- Data Integration
- Customer Segmentation
- KPI Development
- Interactive Dashboard Design

# Data Preparation & Quality Assessment

## Data Cleaning

- ✓ Removed duplicate records
- ✓ Standardized Geography values
- ✓ Corrected invalid salary placeholders
- ✓ Validated missing Customer IDs
- ✓ Preserved non-critical missing surname records

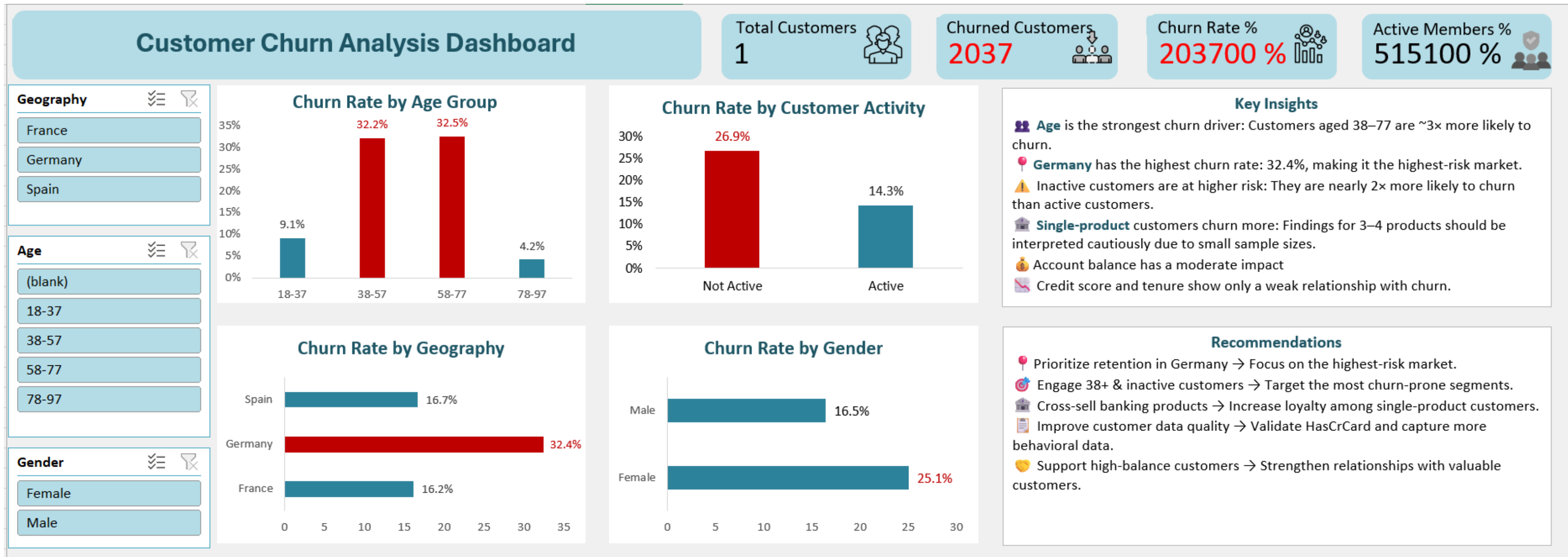
## Data Quality Limitation

- The HasCrCard and IsActiveMember columns contain identical values throughout the dataset.
- This appears to be a source-system issue and limits independent analysis of credit card ownership.



# Dashboard

Interactive dashboard enabling dynamic customer segmentation using Geography, Age Group and Gender filters.

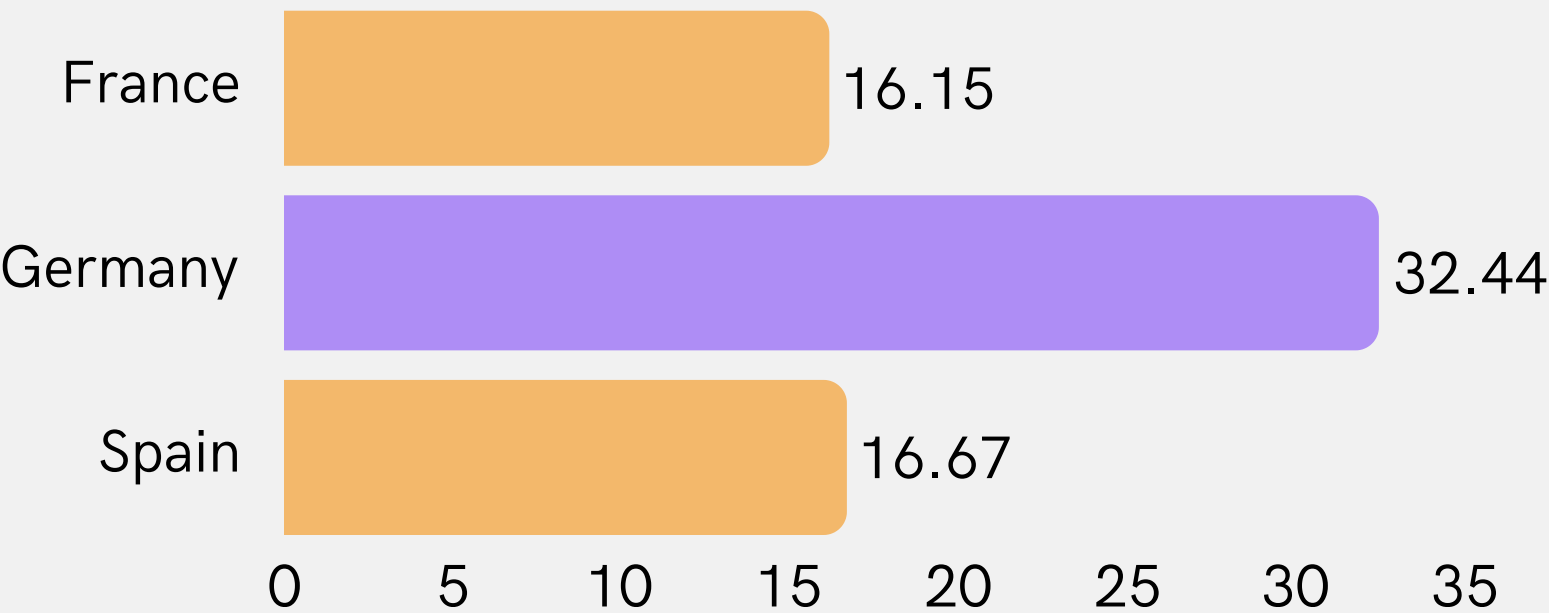


# Executive Insights

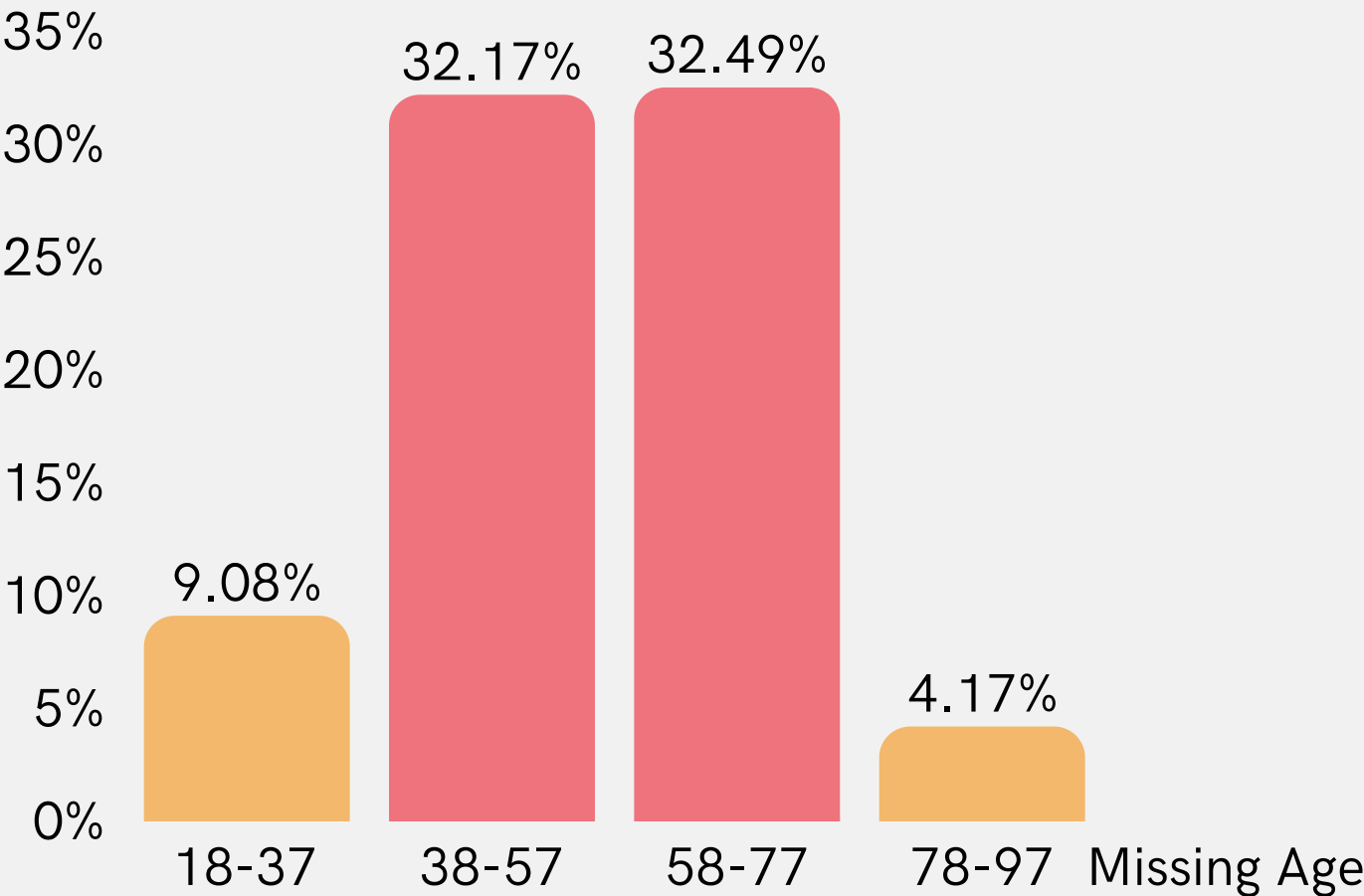
## 1. Germany has the highest churn

Germany records a churn rate of 32.4%, approximately twice that of France and Spain.

Churn Rate by Geography



Churn Rate % by Age Group



## 2. Age is the strongest churn driver

Customers aged 38–77 years exhibit approximately 3× higher churn than younger customers.

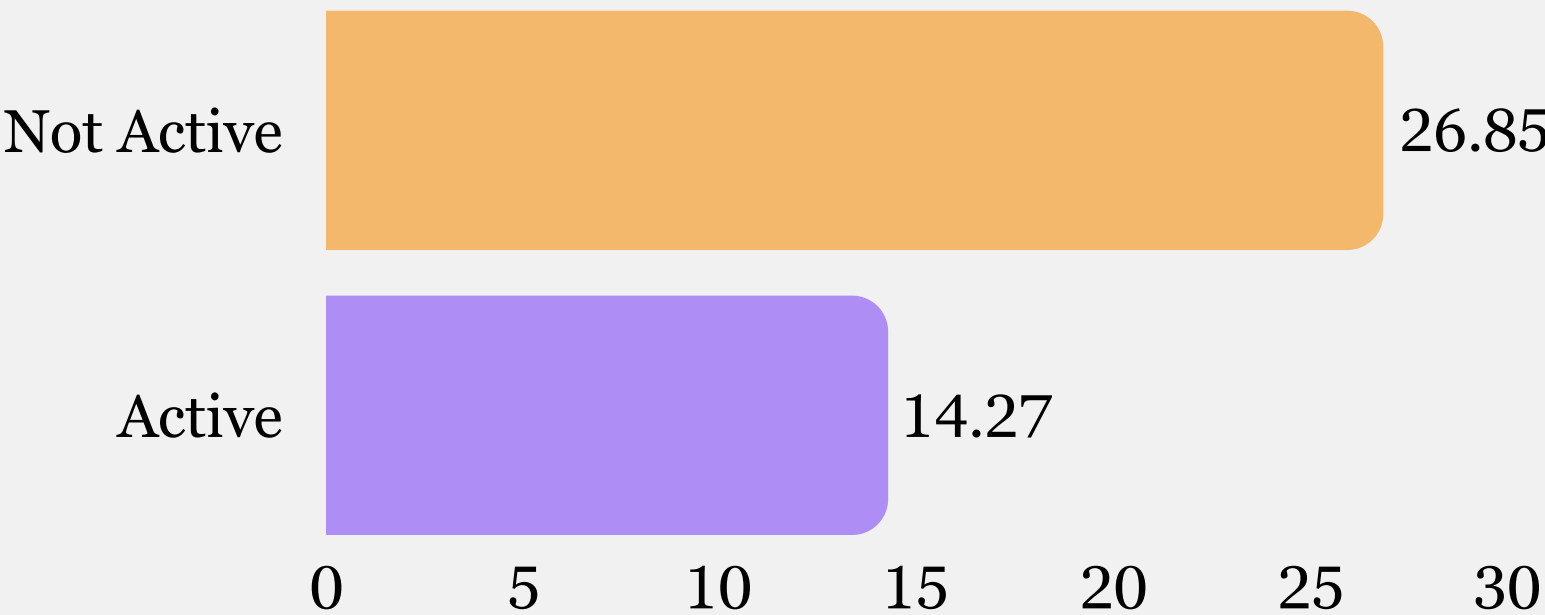


# Executive Insights

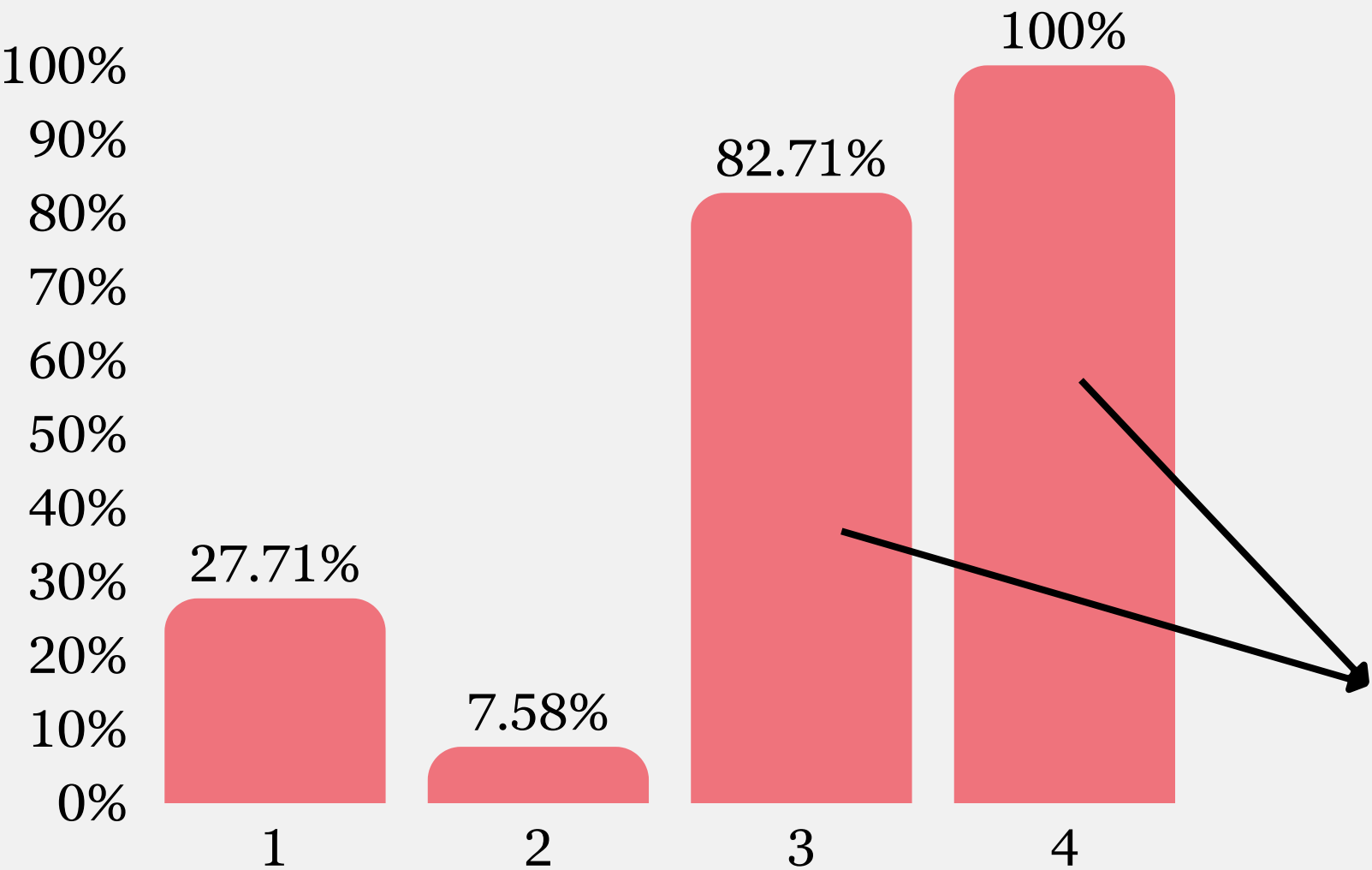
### 3. Customer inactivity significantly increases churn

Inactive customers are nearly 2× more likely to churn.

Churn Rate by Activity



Churn Rate by No. Of Products



### 4. Single-product customers show higher churn

Customers with one banking product exhibit considerably higher churn than customers with two products.

**NOTE :**

3–4 no. of products is based on a very small sample size. Additional data and further investigation are required before drawing business conclusions.

# Executive Insights

## **5. Account balance shows a moderate relationship with churn**

Customers with balances between €100K–€150K demonstrate moderately higher churn, while higher balance groups require additional data for reliable analysis.

## Germany Deeper Analysis

Germany represents the highest-risk market.

Characteristics of high-risk customers include:

- Age above 38 years
- Inactive members
- Single-product customers
- Female customers

This segment should be prioritized for future retention initiatives.

# Business Recommendations

## Immediate

- Prioritize retention campaigns in Germany.
- Target inactive customers with personalized engagement.
- Focus loyalty programs on customers aged 38+.

## Medium-Term

- Promote multi-product adoption through cross-selling.
- Strengthen relationship management for higher-balance customers.

## Long-Term

- Improve source-system data quality.
- Capture behavioural metrics (transactions, digital engagement, complaints) to improve churn prediction.



Thank  
you

Questions & Discussion

Connect with me -

